

Comparative Analysis of ¹³⁷Cs Radiation Attenuation in Aluminum Using MCNP6 Simulations: Influence of Beam Geometry

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1. Introduction

The intensity of ionizing radiation passing through an attenuating medium decreases exponentially according to the Beer-Lambert law, characterized by the linear attenuation coefficient [1]. This study evaluates the influence of pencil-beam and divergent-beam geometries on the determination of the linear and mass attenuation coefficients of aluminum when subjected to 0.662 MeV gamma photons from a ¹³⁷Cs point source. Utilizing the Monte Carlo method (MCNP6) to simulate radiation transport, the results are directly compared against the standard reference database from the National Institute of Standards and Technology (NIST/XCOM) to quantify the influence of beam geometry on the accuracy of the calculated attenuation coefficients[2][3].

2. Methodology

The computational modeling of the attenuation experiment was developed using the MCNP6 code and MCNP Visual Editor. The setup comprised a ¹³⁷Cs point source, a solid aluminum absorber (density 2.6989 g.cm⁻³, varying from 1 to 5 cm in thickness), a detailed NaI(Tl) detector, and a 0.5 cm diameter post-target lead collimator to reduce scattered photons. Two geometries were investigated: an ideal, monodirectional pencil beam, and a divergent beam starting with a 12.92° semi-angle that was systematically halved down to 1.62°. To approximate realistic experimental conditions, the NaI(Tl) detector was modeled including magnesium oxide (MgO) reflectors and a simplified photomultiplier tube (PMT) region made of aluminum. The linear attenuation coefficient was calculated by extracting transmitted intensities and applying a linear regression to the rearranged Beer-Lambert equation: $I = I_0 \cdot e^{-\mu x}$.

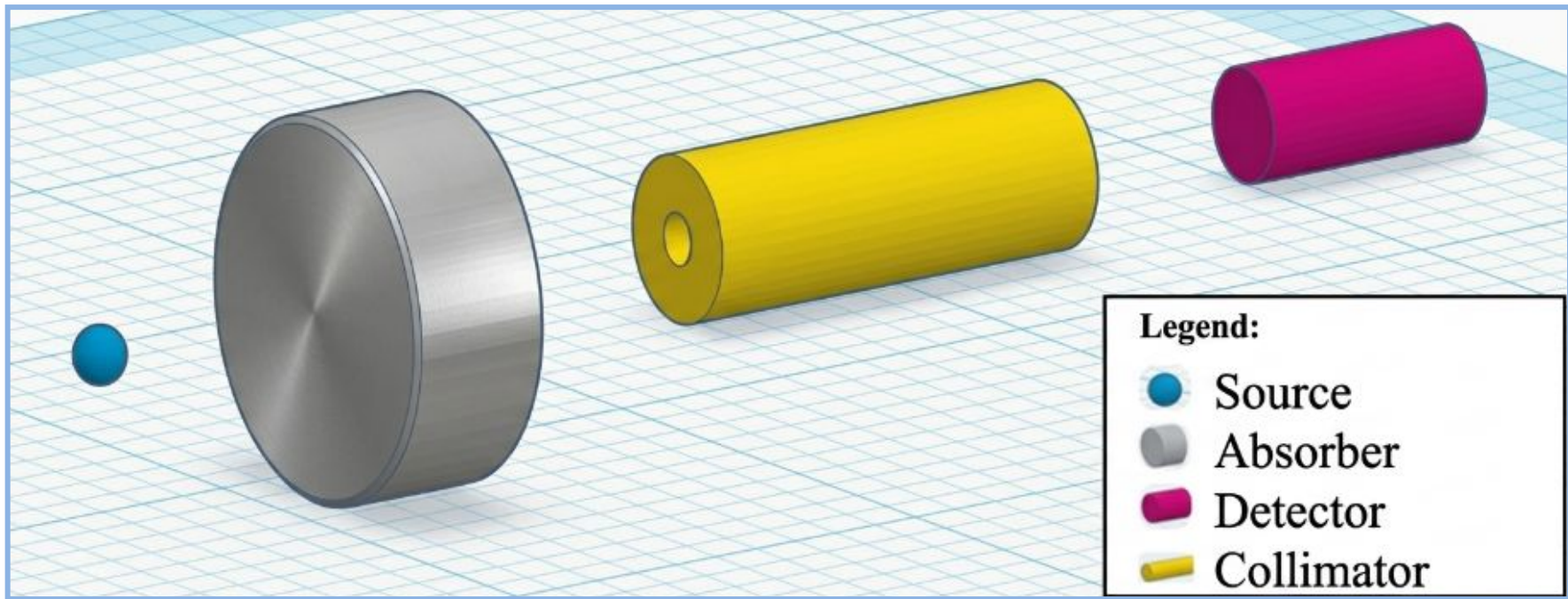


Figure 1: Three-dimensional geometry employed in the MCNP6 attenuation simulations.

3. Results and Discussion

A linear regression analysis of $\ln(I_0/I)$ versus absorber thickness determined the attenuation coefficient for each configuration. Decreasing the divergent beam semi-angle systematically reduced the relative error, demonstrating convergence toward the ideal narrow-beam condition. When compared to the theoretical mass attenuation coefficient from the NIST/XCOM database (7.433×10^{-2} cm².g⁻¹ for aluminum), the pencil beam model demonstrated the highest accuracy, yielding a relative error of only 0.03%.

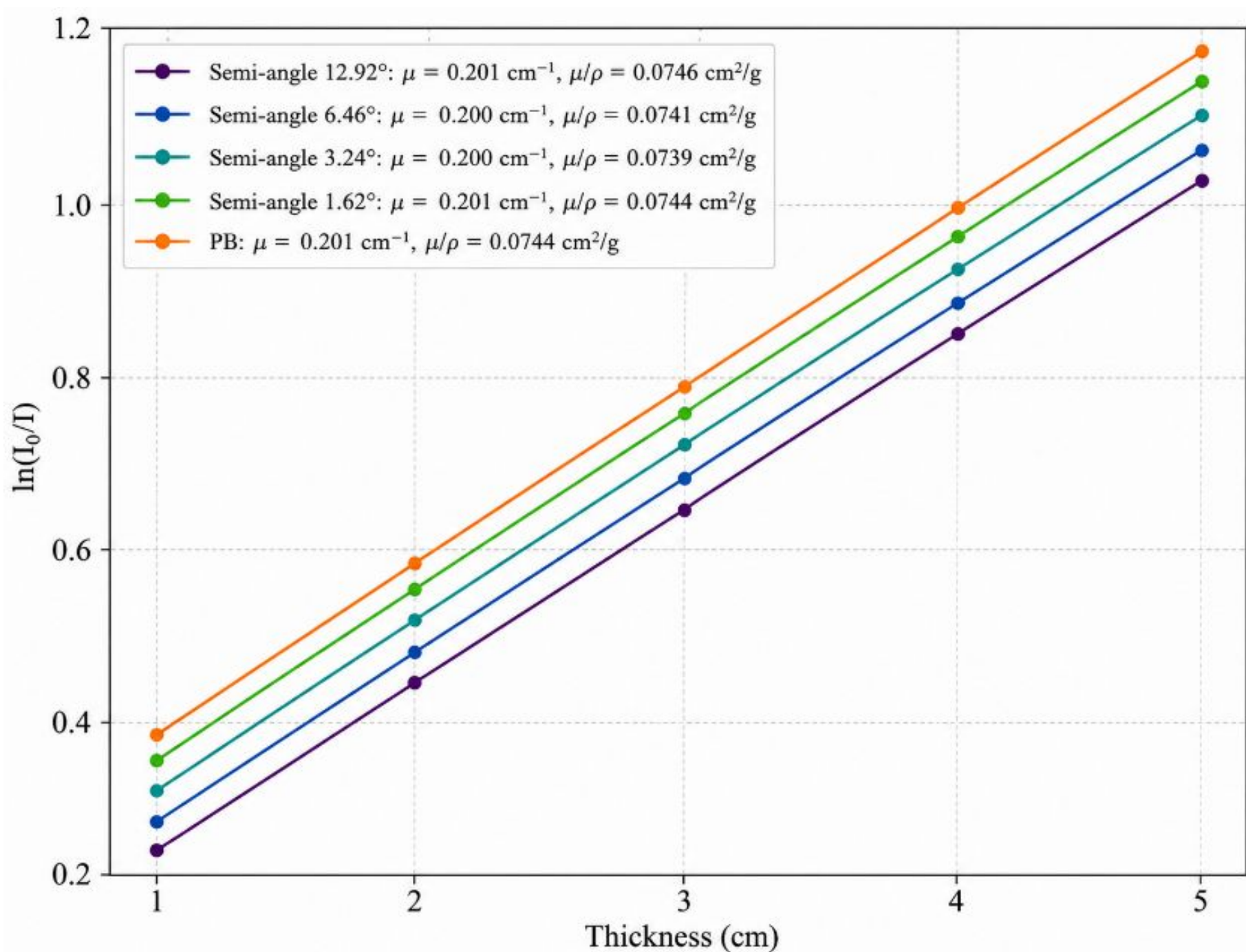


Figure 2: Comparison of attenuation linear regressions obtained for different beam geometries.

Divergent geometries exhibited larger deviations due to geometric divergence — which increases the effective photon path length — and a higher probability of secondary Compton-scattered photons reaching the detector, despite the post-target collimation.

Table I: Comparison of attenuation linear regressions obtained for different beam semi-angles.

Beam Geometry	Semi-angle (°)	μ (cm ⁻¹)	μ/ρ (cm ² .g ⁻¹)	Relative Error (%)
Divergent Beam	12.92	0.201392	0.074620	0.39
	6.46	0.199867	0.074055	0.37
	3.24	0.199562	0.073942	0.52
	1.62	0.200894	0.074435	0.14
Pencil Beam	0	0.200673	0.074354	0.03

4. Conclusions

The results demonstrated that beam geometry significantly influences the determination of attenuation coefficients in Monte Carlo simulations. The pencil beam configuration demonstrated the best agreement with NIST/XCOM reference data, yielding the lowest relative error, whereas divergent beams produced larger deviations due to geometric divergence and scattered radiation. Reducing the semi-angle improved accuracy, emphasizing the necessity of precise beam collimation for reliable measurements. Furthermore, the study validates MCNP6 as a reliable, cost-effective, and safe virtual laboratory for optimizing radiation attenuation experiments.

References

[1] G. F. Knoll, Radiation Detection and Measurement, 4th ed., John Wiley & Sons, Hoboken, USA (2010).
[2] T. Goorley et al., “Features of MCNP6” Annals of Nuclear Energy, vol 87, pp. 772-783, 2016.
[3] “NIST XCOM: Element/Compound/Mixture,” <https://physics.nist.gov/PhysRefData/Xcom/html/xcom1.html> (2026).

